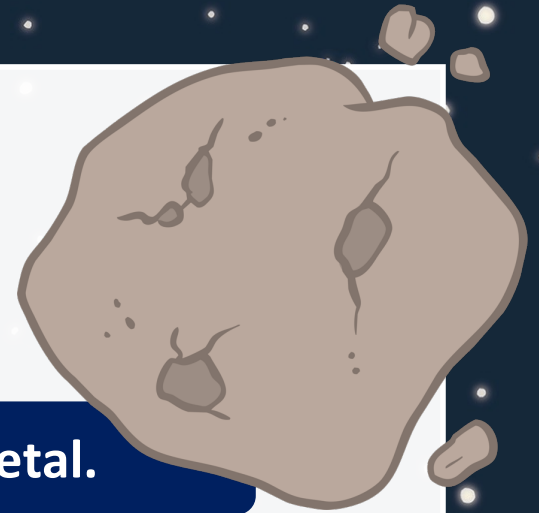


Asteroids, Meteoroids, & Comets



What Is an Asteroid?



Asteroids are made up of carbon, rock, and metal.

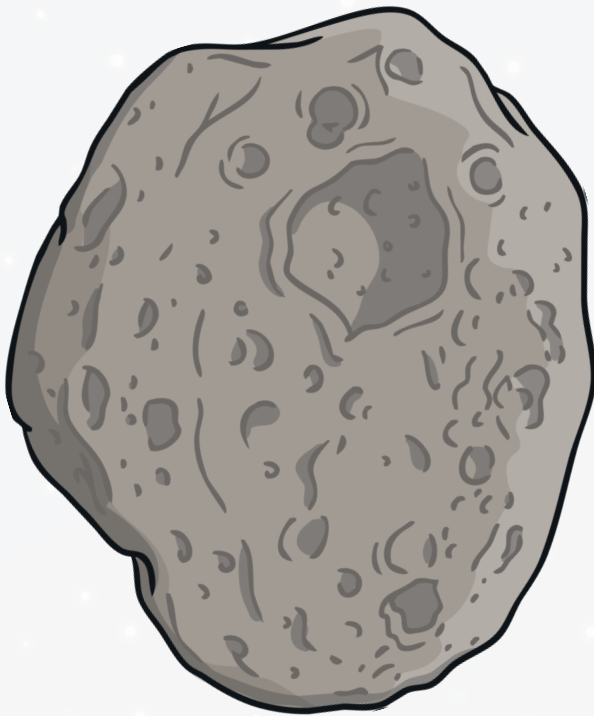
Asteroids revolve around the sun.

Asteroids range in size from 1 meter to more than 900 Km in diameter.

Asteroids are thought to be leftover material from the formation of the solar system which did not form planets.

Asteroids are found mostly between Jupiter and Mars. This is called the asteroid belt.

Asteroid Characteristics



Asteroids are usually irregularly shaped.

Asteroids are larger than pebble-sized meteoroids
Asteroids usually have pits and craters.

Asteroids do not have a tail or coma like a comet.



Think and Share



Think



**Why do you think the asteroid's
surface has craters and pits?**

Share

Share your ideas with a partner.

Famous Asteroid

Hubble image of Ceres, the largest asteroid in the main asteroid belt, compared with four other asteroids and Mars. (Longest dimension for each body in parentheses.)



Image credits: Gaspra, Ida: Galileo (NASA/JPL); Eros: NEAR Shoemaker (JHU/APL).
Vesta and Mars images: HST (NASA/STScI).

Ceres is the largest known asteroid

It orbits the Sun in the Asteroid belt and is large enough to be called a “mini-planet”.

It is thought to contain large amounts of pure water ice below its surface.

Meteors, meteoroids, and Meteorites



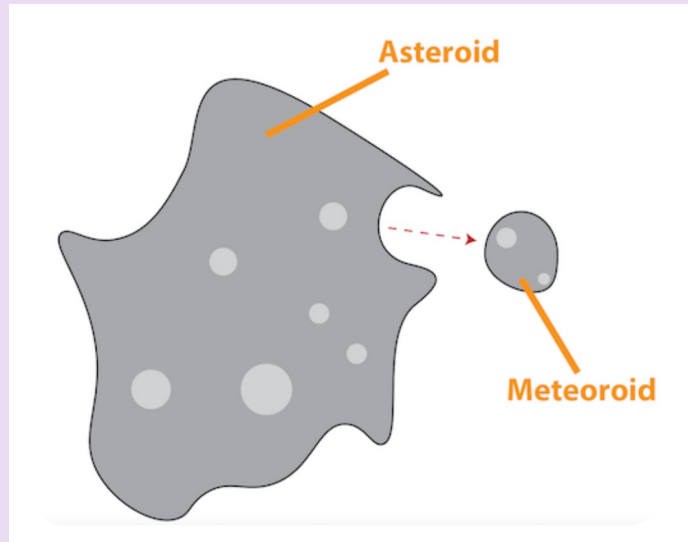
Meteorite hunters look for ancient meteorites in dry places like deserts and Antarctica. Why do you think this is? What tools might you need to find meteorites?

Meteoroid- a small, rocky or metallic body that travels through space, smaller than 1 meter in diameter.

Meteor- a bright streak that results from a meteoroid burning up in Earth's atmosphere, what we call shooting stars.

Meteorite - meteoroid that reaches the Earth's surface without burning up.

From Asteroid to Meteorite



Many meteoroids break off from a larger asteroid.

Large asteroids with a diameter greater than 1 Km only impact Earth about once every 500,000 years.

5 Km objects strike Earth about once every 20,000,000 years.



Meteor Crater in Arizona, USA

The last known impact of an object with a diameter of at least 10 Km happened 66 million years ago.

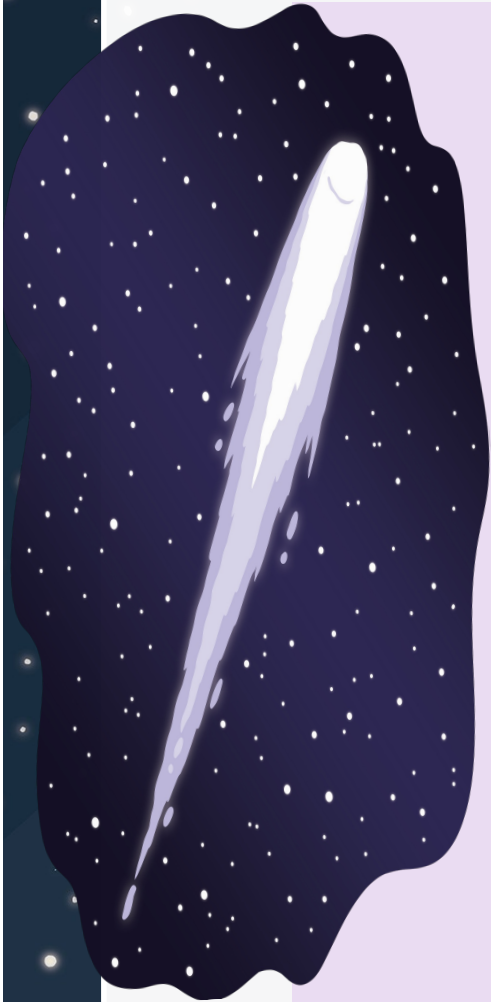
Our atmosphere protects us from most of the damage from impact.

Famous Meteor



The Chelyabinsk meteor was seen over Russia on February 15th, 2013. The object exploded in the atmosphere at a height of about 30 Km. The shock wave injured 1,491 people and damaged around 7,200 buildings across 6 cities. Fortunately, nobody was killed in this event.

Comet Characteristics



Comets are made up of ice and rock, and loosely-packed dust

Because of their composition, scientists sometimes call them "dirty snowballs"

Most comets are round in shape.

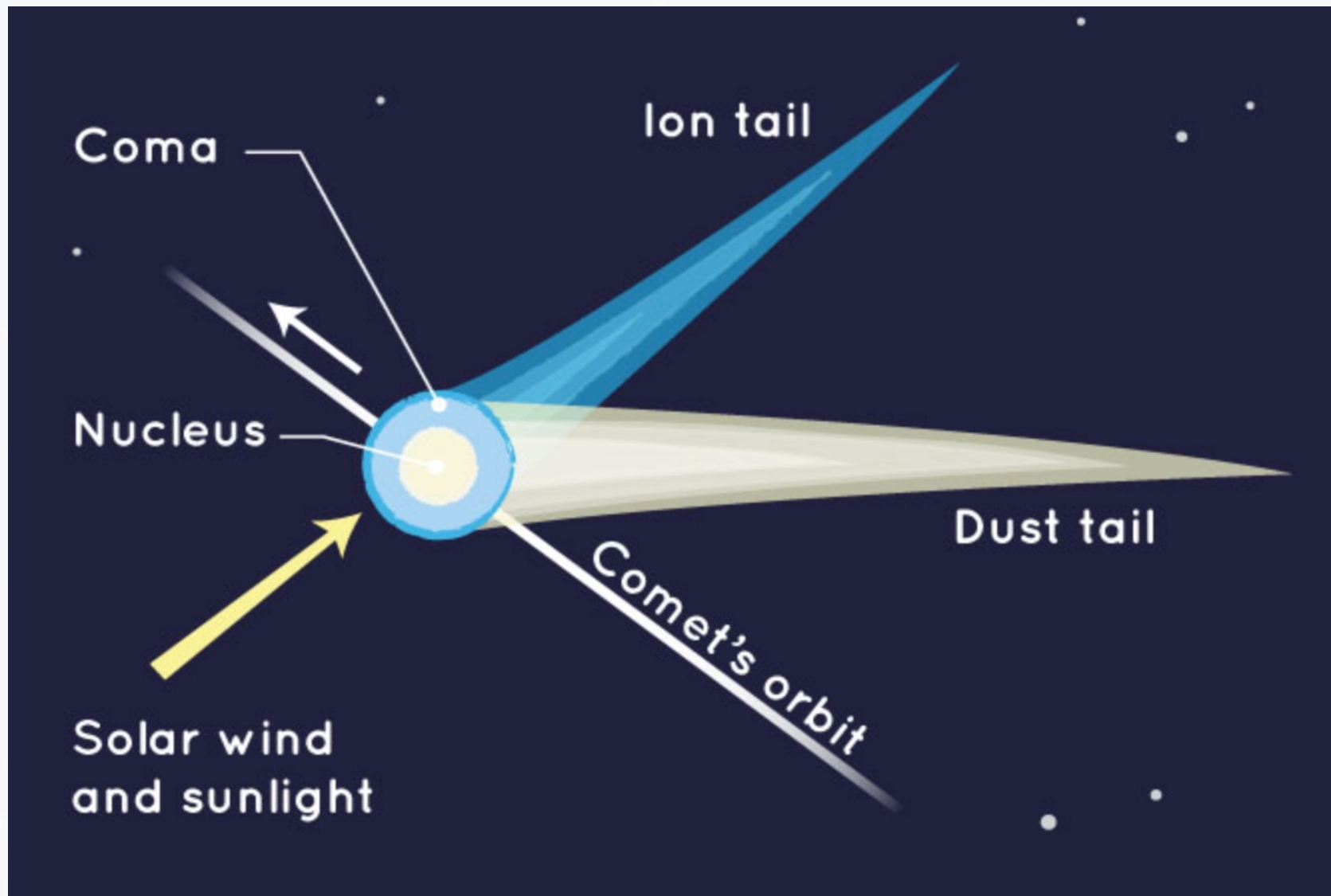
A comet can be a few miles to tens of miles wide.

Comets have a nucleus, coma, and different tails.

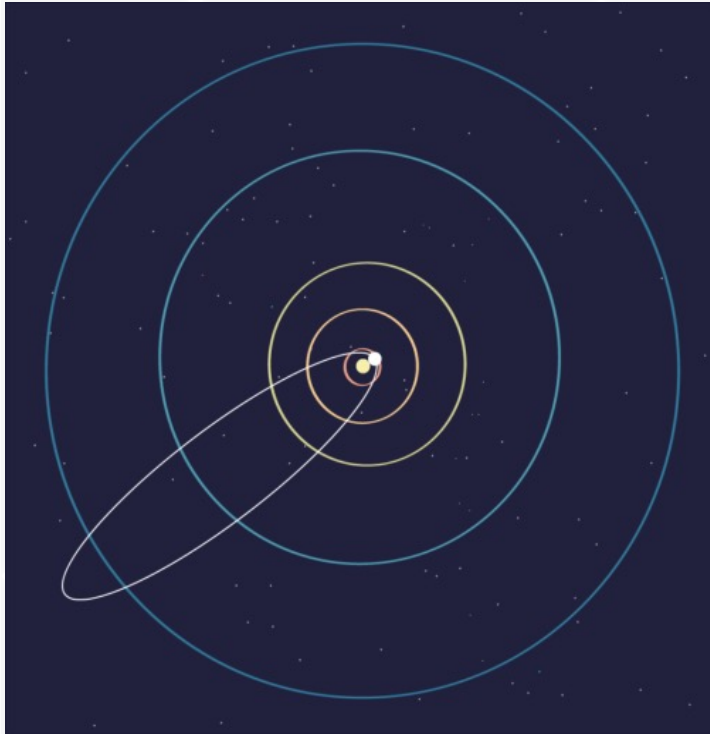
The coma surrounds the center or nucleus of the comet and is made of sublimating ice and dust.

**The tails are made up of ionized gas and dust.
This is the trail the comet leaves behind.**

Parts of a Comet



Comet Orbit

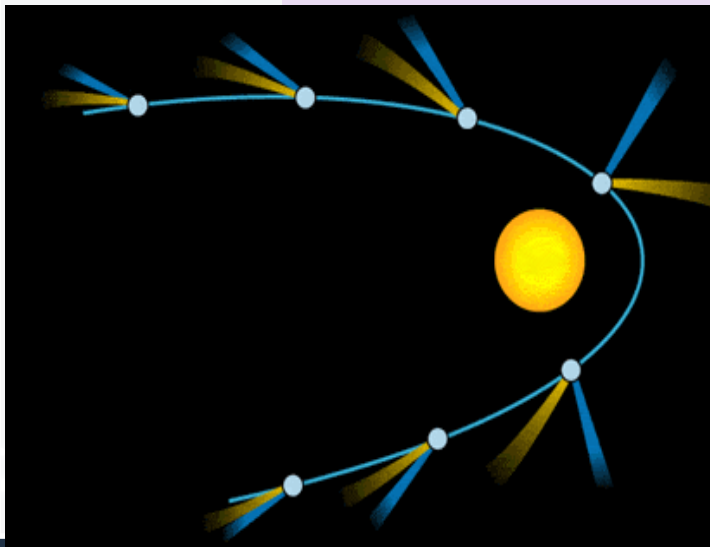


Comets revolve around the Sun in highly elliptical orbits.

A comet's velocity greatly increases as it gets nearer to its star and slows down when it gets further away.

Comets are only bright when they are vaporizing near the Sun. For most of its orbit, it is dark.

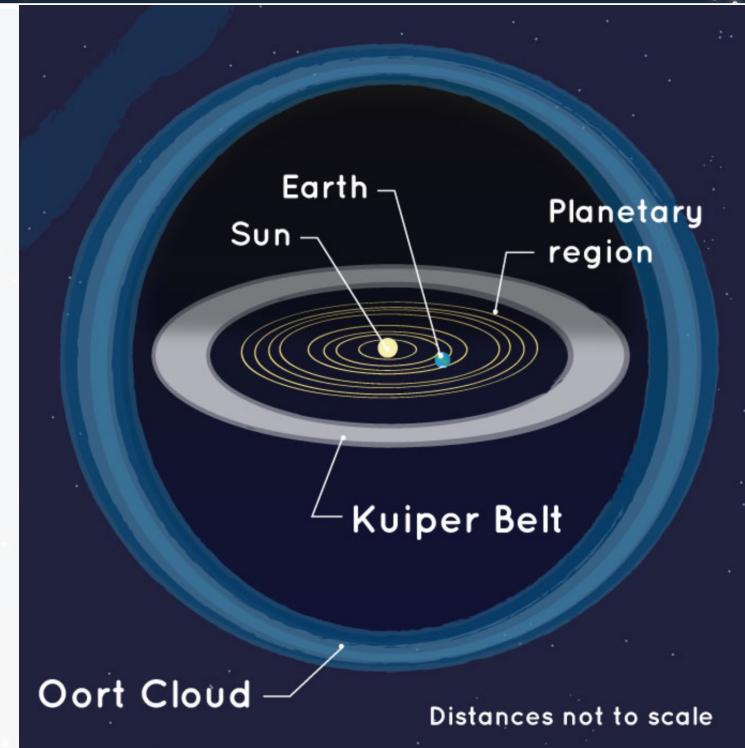
Solar winds always push the ion tail away from the Sun.



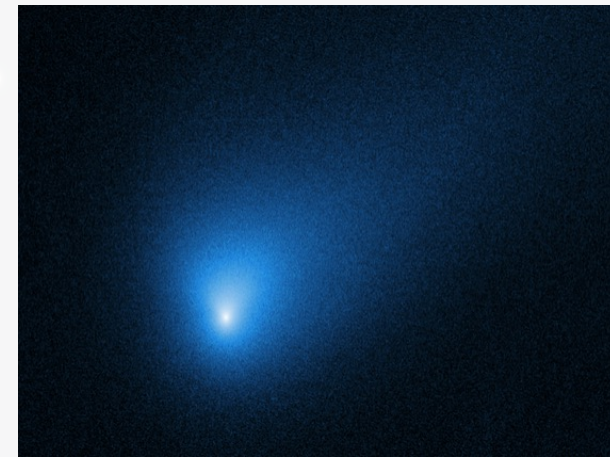
Comet Origins

Comets come from the farthest places in the solar system, the Kuiper Belt and the Oort Cloud.

Only 2 observed objects are known to have come from outside our solar system:

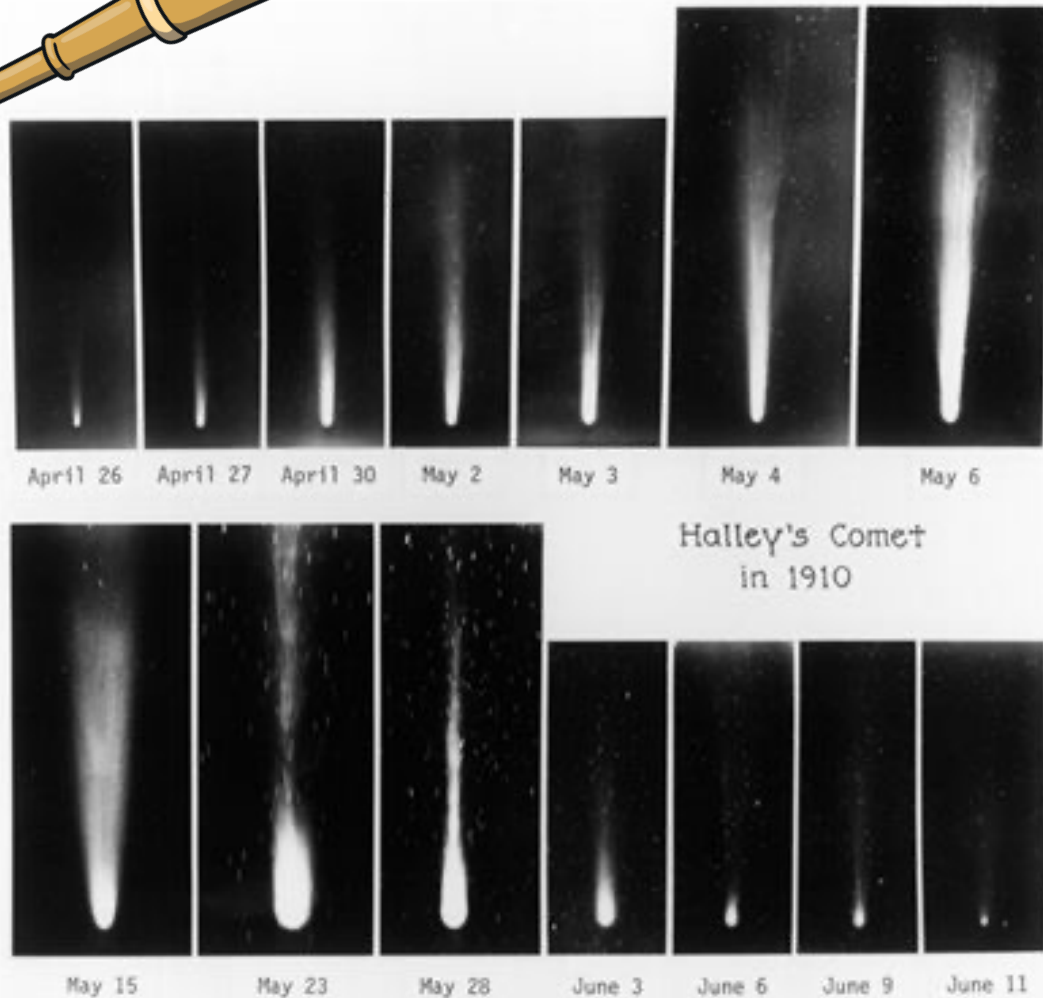


1 Oumuamua
Discovered October 19, 2017



2I Borisov
Discovered August 30, 2019

Famous Comet



Halley's Comet is one of the best-known comets.

It is named after the English astronomer Edmond Halley.

It can be seen from Earth every 76 - 79 years.



Think and Share



Think



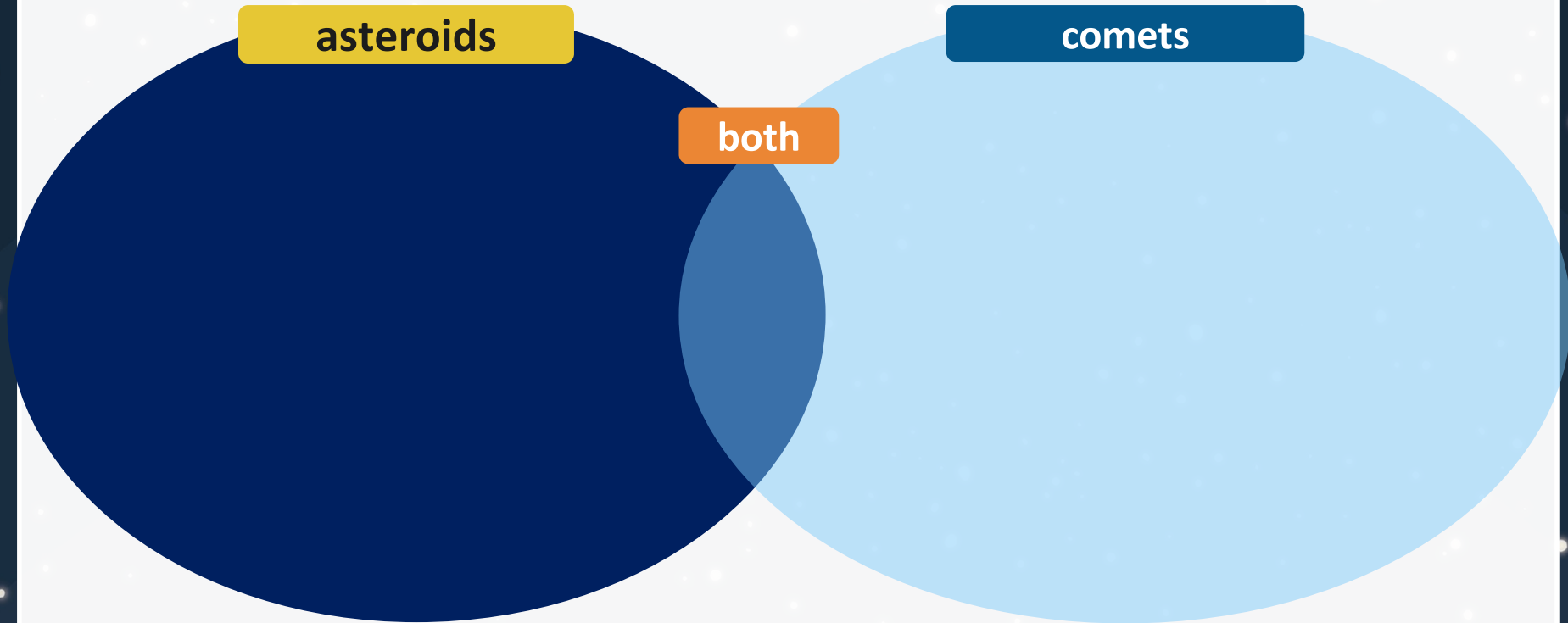
Why do you think the comets leave a trail?

Share

Share your ideas with a partner.

Compare and Contrast

Click and move the words into the corresponding parts of the Venn diagram to compare and contrast asteroids and comets.



Word Bank

orbits sun

have a tail

made of carbon, rock, and metal

found in asteroid belt

visible from Earth

not a planet

have pits and craters

have no tail



Show What You Know!

**What do asteroids and
comets orbit?**

sun

moon



Show What You Know!

**What do asteroids and
comets orbit?**

sun

moon



Try again!

Try again

Next



Show What You Know!

**What do asteroids and
comets orbit?**

sun

moon



Correct

Next



Show What You Know!

An asteroid has a tail.

true

false



Show What You Know!

An asteroid has a tail.

true

false



Try again!

Try again

Next



Show What You Know!

An asteroid has a tail.

true

false


Correct

Next



Show What You Know!

**What is made up of ice,
rocks, and dust?**

asteroid

comet



Show What You Know!

**What is made up of ice,
rocks, and dust?**

asteroid

comet



Try again!

Try again



Show What You Know!

**What is made up of ice,
rocks, and dust?**

asteroid

comet

Correct

Finish



Image credit: NASA/JPL-Caltech/UMD